Preliminaries

Most people meet linear algebra as a way of computing: row reduction, determinants, maybe eigenvalues. This book treats it as a subject you **prove**. Every statement comes with a reason, and those reasons are built from a small set of ideas: sets, functions, equivalence relations, fields, polynomials, and matrices. This chapter collects that toolkit in one place.

You have probably seen some of this before. Some of it, such as negating a statement with several quantifiers, or checking that a rule on equivalence classes is well defined, is exactly where careful readers later stumble. So each section spends its time on the delicate points and moves quickly past the rest.

How to use this chapter. Skim the sections you know, but try the *A. Check your understanding* exercises at the end of each one. If you can do them without looking back, move on. If not, that section is worth a slower read.

What you need. Nothing from this book. School algebra is enough, plus some familiarity with numbers such as $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$. One convention to note now: in this book the natural numbers $\mathbb{N}$ start at 0.

0.-1.1 Roadmap

- **Statements, implications, and quantifiers.** How mathematical statements are built, and how to negate them correctly.
- **How proofs are built.** Direct proof, contrapositive, contradiction, cases, uniqueness, counterexamples and induction, and the conventions this book's proofs follow.
- **Sets.** Subsets, operations, indexed families, and the habit of proving two sets equal by double inclusion.
- **Functions.** Injective, surjective, and bijective maps; composition; left, right, and two-sided inverses.
- **Relations, equivalence classes, and quotients.** Sorting a set into classes, and defining functions on classes without ambiguity.
- **Complex numbers.** Arithmetic, conjugates, the modulus, polar form, and roots of unity.
- **Fields.** The number systems in which linear algebra works, from $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ to the finite fields $\mathbb{F}_p$.
- **Polynomials.** Degree, evaluation, and why a polynomial is not the same thing as a function.
- **Matrices.** Addition, the three ways to read a matrix product, transpose, trace, and inverses.
- **Groups and permutations.** One structure shared by invertible matrices, the nonzero elements of a field, and rearrangements of a set.

0.-1.2 Named moves

These proof techniques appear throughout the book. Each is introduced in this chapter.

- **Contrapositive.** To prove "if P then Q", prove "if not Q then not P".
- **Assume two.** To prove something is unique, suppose two objects have the property and show they are equal.
- **Double inclusion.** To prove $A = B$, prove $A \subseteq B$ and $B \subseteq A$.
- **Suppose $f(a) = f(b)$.** The standard first line when proving that a function is injective.
- **Check well-definedness.** A rule given through a representative must not depend on

which representative is chosen.

- **Induction on n.** Find a smaller instance of the problem inside the larger one.